

Adrin Alias

Engineering Portfolio

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 University of Texas at Arlington

 B.S. Mechanical Engineering, December 2025

Brake Dyno

Sensors & Data Acquisition

Firmware Development

Signal Conditioning

Python

Circuit Design

CAN Communication

SolidWorks (CAD)

What & Why

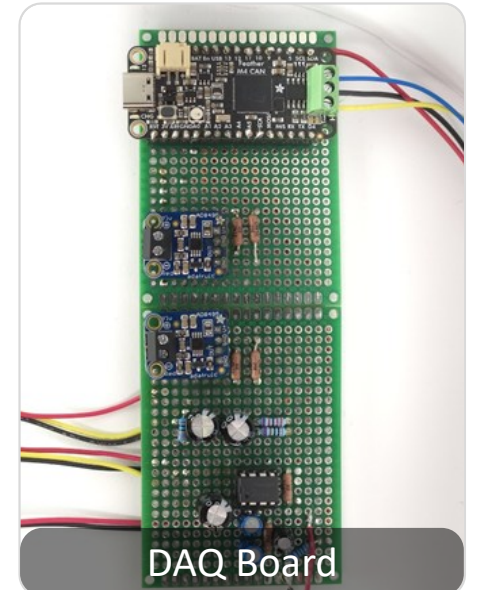
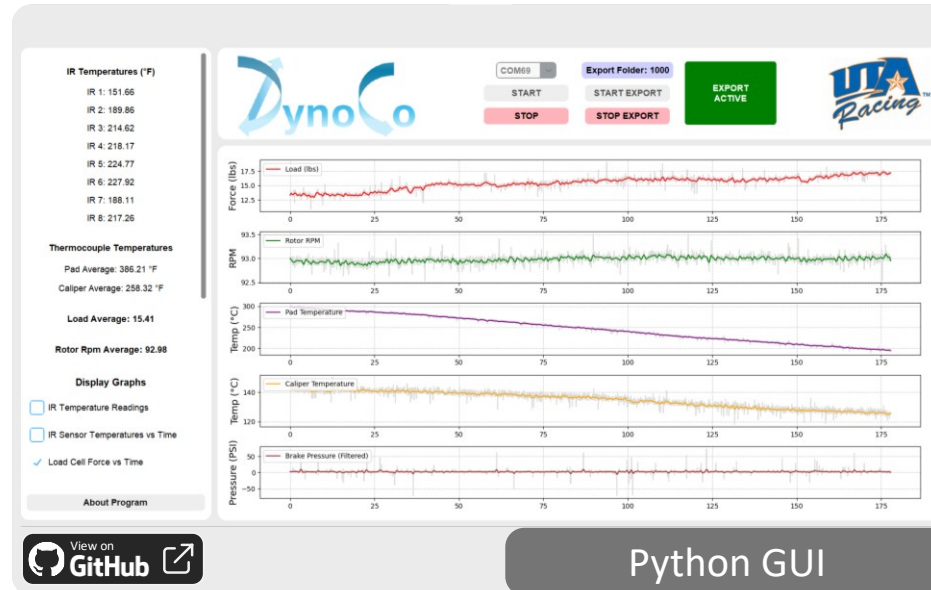
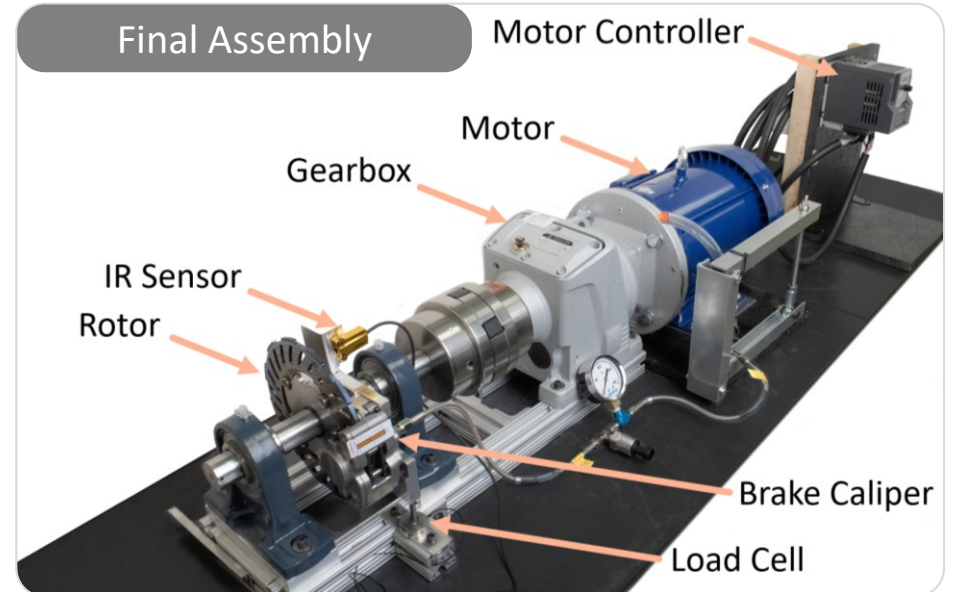
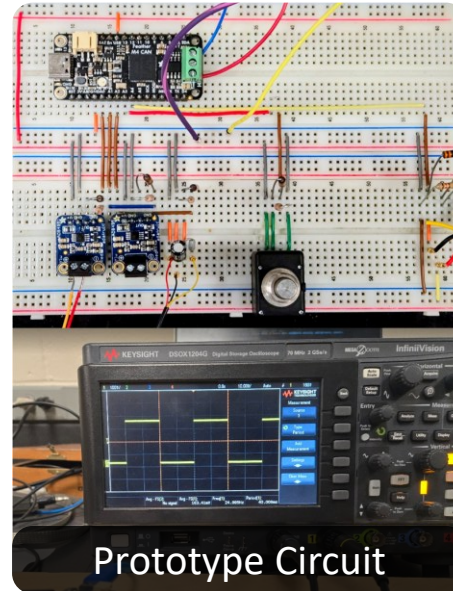
Developed a Brake Dynamometer (6-person team) for the Formula SAE team to measure the coefficient of friction for various brake components, enabling data-driven design decisions to improve lap times. I was responsible for the complete development of its custom DAQ system, including the hardware, firmware and GUI software.

How

The test rig uses a motor to spin a brake assembly while measuring torque, speed, and temperature. I developed the DAQ system using an SAME51 MCU, which involved designing the **signal conditioning circuits** (voltage dividers, RC filters) and programming the **C++ firmware** to process all sensor data, manage timing, and transmit it via Serial. I also created a **Python GUI** for real-time data plotting & logging.

Results

- Delivered a functional DAQ with **20 Hz** real-time processing and logging, producing a coefficient of friction vs. temperature graph that **validated manufacturer claims**.
- Reduced system cost by **94%** (from \$8200 to \$500) compared to a commercial NI DAQ by focusing on core requirements.
- Awarded 2nd Place at UTA Innovation Day 2025. [🔗](#)



IOT Smart Binds

May 2024 – August 2024

Powered by ESP32

SolidWorks (CAD)

3D Printing

Enclosure Design

Rapid Prototyping

Firmware Development

Testing & Validation

What & Why

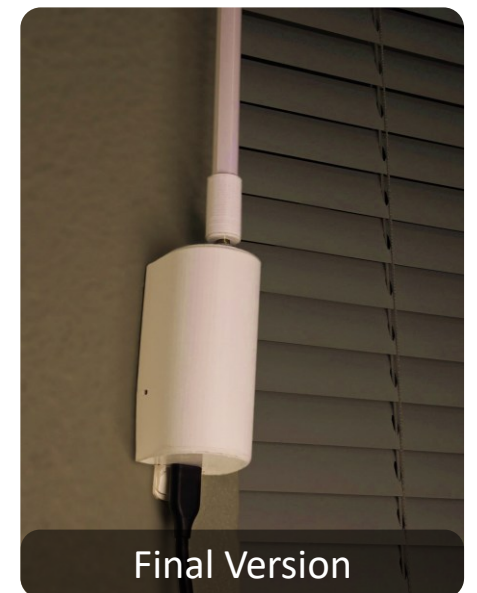
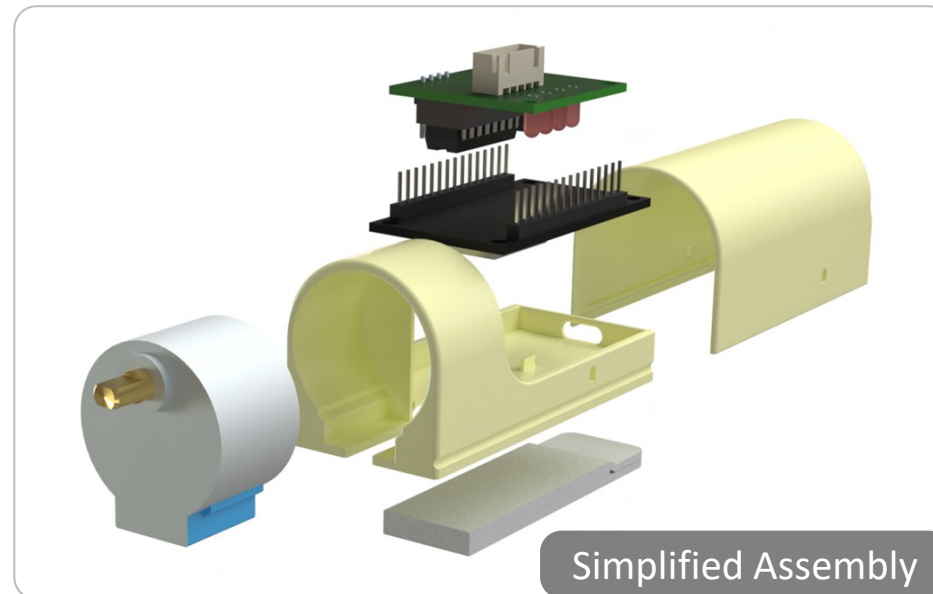
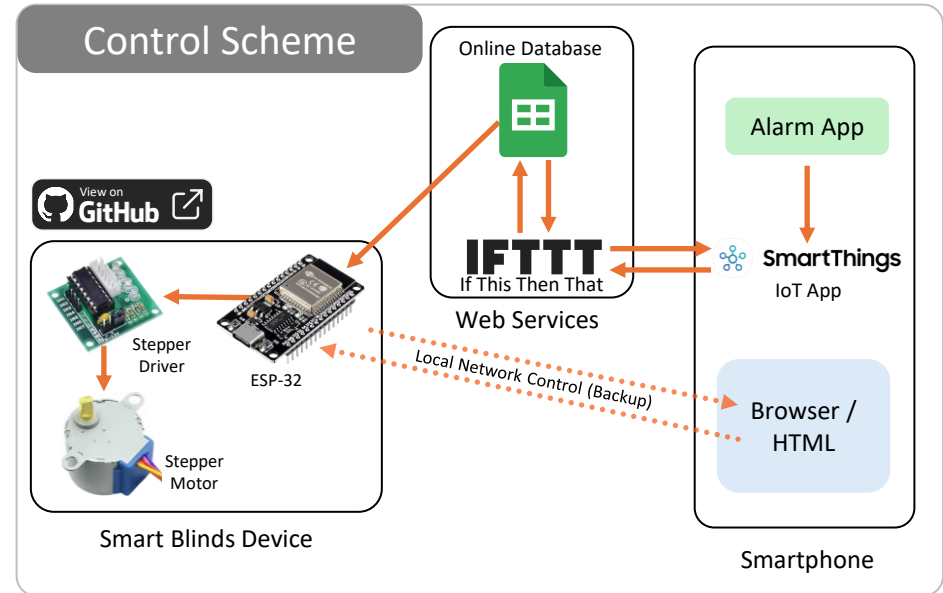
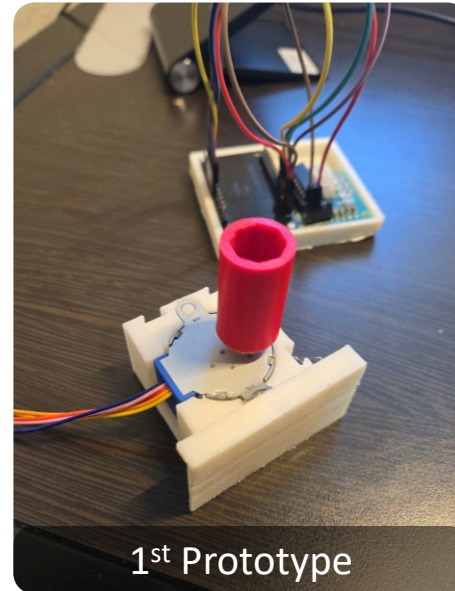
- A small electromechanical device to automate opening & closing blinds at a lower cost than commercial alternatives.
- Personal project to improve sleep quality and quantity by only letting in light from wake-up time till dusk.

How

- Utilized an ESP32 microcontroller and an off-the-shelf stepper motor driver. Programmed the device in C++ (Arduino) for automation and open-loop motor control.
- The firmware connects via Wi-Fi, fetches daily sunset times from a public weather API and syncs with an NTP server for timekeeping. Samsung SmartThings is used to detect the morning alarm.

Results

- Successfully created a **fully autonomous system** that reliably opens and closes daily based on user alarm info and daily sunset times.
- **85% lower hardware cost** than similar alternatives in the market. (\$70 → \$10)
- Completed a 30-day continuous testing with **100% reliability** in opening and closing.
- *Future Work:* Upgrading system to use an H-bridge and a car power window motor to open bigger blinds with strings.



Air Heaters

Enclosures & Control Systems

SolidWorks (CAD)

Digital Twin

Electrical Wiring

Manufacturing

Sheet Metal Design

Circuit Design

What & Why

- Part of a **high temperature test rig** to simulate the thermodynamic state of a jet engine turbine that needs temperatures up to **750 °C**, net air flowrate of **85 SCFM** at pressures up to 60 psi.
- Required a custom solution because there was no off-the-shelf solution. (Price info unavailable)
- I was responsible for the design of the heater enclosures & the design & integration of the power & control systems.

How

- Designed the SS 304 enclosures in SolidWorks, applying DFMA principles to streamline production with external fabrication shops.
- Collaborated with the university machine shop for manufacturing and welding.
- Personally assembled, tested, and wired all four heater modules, integrating them with an off-the-shelf PID controller to create a complete, functional system.

Results

- Successfully implemented an **8.4 kW air heater system** across 4 heaters and achieved the target temperatures and flow rates.
- Reduced manufacturing cost 30%** by outsourcing sheet metal fabrication.

Content shared with permission | TESLAB @ UT Arlington | June 2024 – October 2024

